Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

Nigel Li Yutong Han

Columbia University, New York

Abstract

Reinforcement learning (RL) portfolio allocators are often given a tail-risk limit through a Lagrangian penalty on Conditional Value-at-Risk (CVaR). We show two ways in which this limit fails with no visible symptom. When the CVaR cost is on a far smaller scale than the reward, the penalty weight λ stays at zero. A single stress test can also reverse how the learner ranks against classical optimisers. We scale the dual step to the CVaR excess and add a decision-time filter that moves each over-budget portfolio toward minimum variance before it trades. The filter cut the out-of-sample breach rate from **75.1% to 7.4%**. Over 40 walk-forward folds on two ETF universes, the constraint lowered CVaR_{99} in **26 of 27** untied folds ($p = 8.8 \times 10^{-6}$).

Motivation

In the week of 16 March 2020 the S&P 500 fell about 15% and the VIX closed at 82.69. An RL allocator trained only for return kept its equities. The same allocator trained with a CVaR budget had already moved into Treasuries and cash, and its drawdown was 7.7% against 16.3% (Fig. 1).

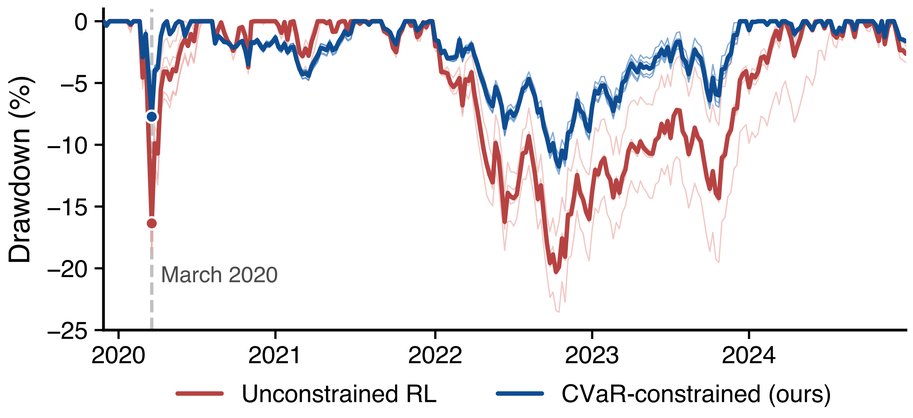


Figure 1. Drawdown over 2020 to 2024. Thick lines are five-seed means and thin lines individual seeds.

Problem and measures

CVaR_α is the average loss in the worst $(1-\alpha)$ share of weeks [1]. With $\alpha = 0.95$ it averages the worst 5%, so it measures how large the losses are beyond the Value-at-Risk (VaR) threshold (Fig. 2).

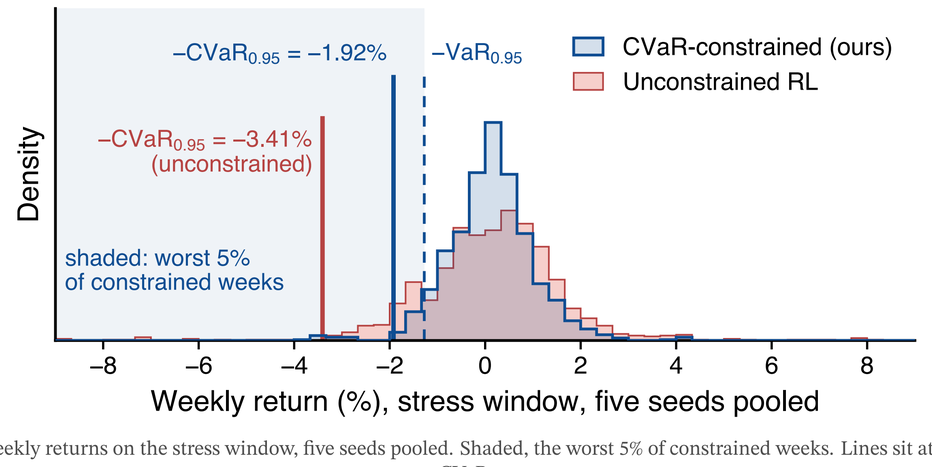


Figure 2. Weekly returns on the stress window, five seeds pooled. Shaded, the worst 5% of constrained weeks. Lines sit at minus each CVaR.

Each week the allocator chooses long-only weights w_t on seven ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL), using 887 weekly returns from 2008 to 2024. The problem is a constrained Markov decision process [2],

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad \mathrm{CVaR}_{0.95}(L) \le d = 1.2\%, \qquad \mathrm{CVaR}_\alpha(L) = \min_{\zeta \in \mathbb{R}} \Big\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}\big[(L-\zeta)^+\big] \Big\}.$$

A test week is a breach when the trailing 52-week $\mathrm{CVaR}_{0.95}$ exceeds d, so the breach rate over the test weeks $\mathcal{T}$ is

$$B = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbf{1}\big[\widehat{\mathrm{CVaR}}_{0.95}(r_{t-52:t-1}) > d\big],$$

with 5% of weeks tolerated. Sharpe ratios are annualised from weekly net returns, and their significance is judged with the Probabilistic and Deflated Sharpe ratios [4].

Failure 1. The penalty weight never moves

With a dual step $\eta_\lambda = 10^{-3}$, λ stayed near zero for all 1500 training updates (Fig. 3). The CVaR excess is about 10^{-2} per week, so each update moved λ by about 10^{-5}. With $\eta_\lambda = 5.0$, λ rose to between 0.1 and 0.3. A model-free actor-critic failed in the same way when its cost signal (about 10^{-3}) was left unstandardised next to a unit-scale reward signal.

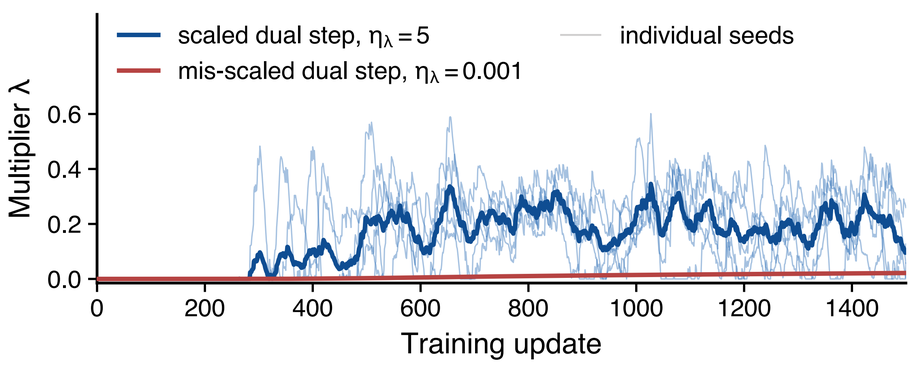


Figure 3. The penalty weight λ during training, five seeds.

Tail metrics do not reveal the failure. Even with λ near zero, the rest of the pipeline (the anchor, the projection and checkpoint selection under the CVaR limit) lowered CVaR_{99} by 61% against the unconstrained learner (0.025 against 0.065, permutation test $p = 0.0044$).

Remedy. Monitor λ during training, since a multiplier that stays at zero means the constraint is inactive, and set the dual step to the scale of the CVaR excess. With $\eta_\lambda = 5.0$ the penalty binds in training, although the portfolio still breaches its budget out of sample, which the decision-time filter addresses.

Failure 2. The test protocol reverses the ranking

A walk-forward test retrains the model on a rolling window and evaluates it on the next unseen block. On a 31-asset universe, one fixed stress split ranked the constrained learner above minimum variance (Sharpe 0.85 against 0.23). A 20-fold walk-forward reversed the order (0.74 against 1.40, Fig. 4). With Ledoit-Wolf shrinkage [5], minimum variance still reached only 0.25 on the split. The tail-risk reduction against unconstrained RL held under both tests.

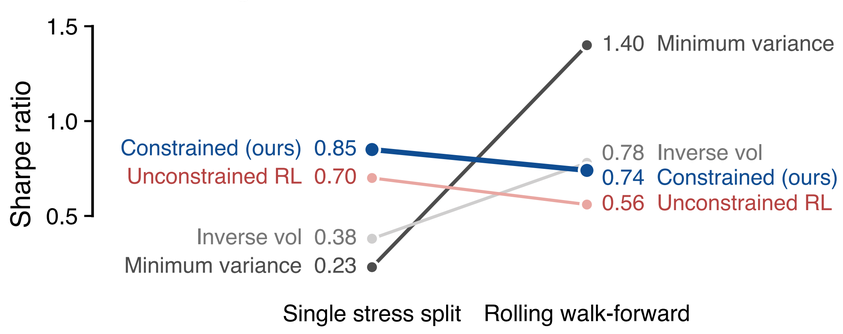


Figure 4. The same four methods ranked by two tests, 31-asset universe.

Remedy. Compare learned allocators with optimisers on rolling walk-forward folds, with multiple-testing control and the Deflated Sharpe ratio [4]. We report walk-forward results alongside the stress window throughout.

Our method

We train the allocator end to end with a CVaR penalty whose dual step matches the scale of the CVaR excess, and a filter checks every portfolio before it trades. Figure 5 shows the weekly decision as four maps, defined in steps (i) to (iv).

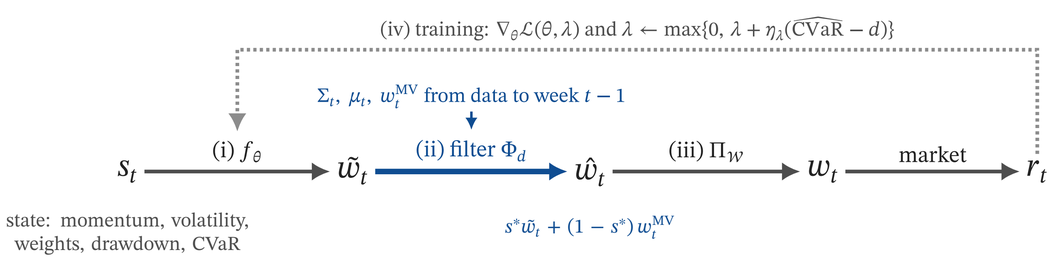


Figure 5. The decision at week t as a composition of maps.

(i) *Policy.* An MLP f_θ tilts an adaptive inverse-volatility anchor a_t, $\tilde{w}_t = \mathrm{softmax}(\log a_t + f_\theta(s_t))$. At initialisation $f_\theta \approx 0$, so training starts from the anchor and learns a state-dependent tilt.

(ii) *Filter.* From the trailing 104 weekly returns it forecasts $\widehat{\mathrm{CVaR}}(w) = c_\alpha \sigma_p(w) - \mu_p(w)$, with $c_\alpha = \varphi(\Phi^{-1}(\alpha))/(1-\alpha)$ and $c_{0.95} = 2.06$. An over-budget proposal is mixed with the trailing minimum-variance portfolio w_t^{MV}, keeping as much of the proposal as the budget allows, $s^* = \max\{s \in [0,1] : \widehat{\mathrm{CVaR}}(s\tilde{w}_t + (1-s)w_t^{\mathrm{MV}}) \le d\}$. It is a safety layer in the sense of [3] and uses data up to week $t-1$ only.

(iii) *Projection.* $\Pi_{\mathcal{W}}$ enforces long-only weights, $w_i \le 0.40$ and $\|w_t - w_{t-1}\|_1 \le 0.50$. The market then pays r_t net of a 5 bps cost per unit of turnover.

(iv) *Training.* On random windows of past weeks we back-propagate

$$\mathcal{L}(\theta, \lambda) = -\mathrm{Sharpe}(r) + \lambda\, \mathrm{ReLU}\big(\widehat{\mathrm{CVaR}} - d\big) + \kappa\, \mathrm{turnover}, \qquad \lambda \leftarrow \max\{0,\ \lambda + \eta_\lambda(\widehat{\mathrm{CVaR}} - d)\},$$

where $\widehat{\mathrm{CVaR}}$ is the mean of the worst 5% of window returns, so its gradient reaches only tail weeks. We keep the checkpoint with the best validation objective among those within the validation CVaR limit.

Table 1. Training and evaluation details.

component	setting
budget	$\alpha = 0.95$, $d = 1.2\%$ weekly CVaR, up to 5% of test weeks may breach
dual step	$\eta_\lambda = 0.001$ (mis-scaled) or 5.0 (scaled)
training	1500 updates per ablation arm and 2500 in the stress study, seeds 7, 13, 23, 42, 2025
stress test	train 2008 to 2018, test the 267 weeks from late 2019 to 2024
walk-forward	train 312, validate 52, test 26 weeks, step 26, 3-seed ensemble per fold
classical baselines	equal weight, inverse volatility, minimum variance, mean-variance, risk parity, min-CVaR LP, refit weekly on the trailing 104 weeks (52 for inverse volatility)

Decision-time filter

A larger dual step made λ move, but the portfolio was then over budget in 75.1% of test weeks. Classical rules met the same budget in 92% of weeks, so the budget was attainable. Adding the filter lowered the breach rate to **7.4%** (Table 2). Raising λ during testing without the filter left the breach rate at 81.9%, which attributes the improvement to the filter.

Table 2. Stress window, five-seed means. Breach is the share of test weeks over budget, with 5% tolerated.

variant	Sharpe	$\mathrm{CVaR}_{0.99}$	breach
no constraint	0.626	6.5%	100.0%
penalty, $\eta_\lambda = 0.001$	0.987	2.5%	7.9%
penalty, $\eta_\lambda = 5.0$	0.911	2.8%	75.1%
penalty, λ raised during testing	0.826	3.1%	81.9%
penalty, $\eta_\lambda = 5.0$, with filter	**0.851**	**1.6%**	**7.4%**
minimum variance	0.901	4.4%	7.9%

Tail risk across test settings

The walk-forward test retrains every 26 weeks on the previous six years and tests on the next 26 unseen weeks, giving 20 folds per universe from 2015 to 2024. In Figure 6 each fold is one point. Below the diagonal the constraint lowered $\mathrm{CVaR}_{0.99}$. On it the budget never bound, so both models held the same portfolio.

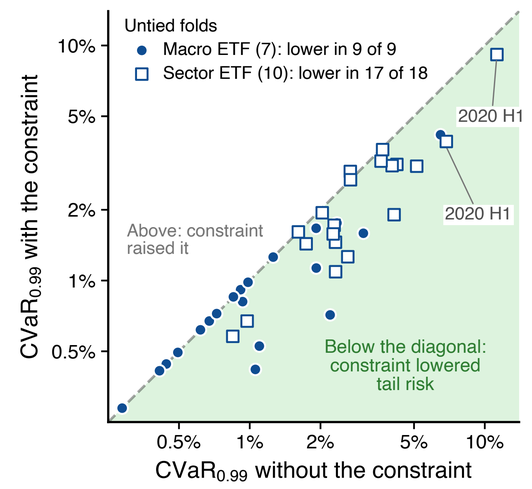


Figure 6. Walk-forward folds, $\mathrm{CVaR}_{0.99}$ without against with the constraint (log scale). The largest reductions came in the first half of 2020.

Table 3. Tail risk with and without the constraint in every test setting.

setting	test periods	$\mathrm{CVaR}_{0.99}$ without → with	evidence
stress window, 7 ETFs	267 weeks, 5 seeds	6.47% → 3.27%	95% BCa CI [−3.5, −2.2] pp
stress window, 10 sector ETFs	267 weeks, 3 seeds	10.9% → 7.6%	95% BCa CI [−5.0, −3.2] pp
walk-forward, 7 ETFs	20 folds	1.43% → 1.02%	lower in 9 of 9, $p = 0.0077$
walk-forward, 10 sector ETFs	20 folds	3.36% → 2.50%	lower in 17 of 18, $p = 0.0003$
both universes pooled	**40 folds**		**lower in 26 of 27, $p = 8.8 \times 10^{-6}$**
walk-forward, 31 assets	20 folds		$p = 6 \times 10^{-4}$

Fold counts exclude tied folds. Each p is a Wilcoxon signed-rank test on $\Delta_k = \mathrm{CVaR}^{\mathrm{con}}_{0.99,k} - \mathrm{CVaR}^{\mathrm{unc}}_{0.99,k}$, with $W^+ = \sum_{k:\,\Delta_k \ne 0} R_k\, \mathbf{1}[\Delta_k > 0]$ and R_k the rank of $|\Delta_k|$. Both universes pass a Benjamini-Hochberg correction [6], and clustering folds by calendar gives $p = 3.3{\times}10^{-4}$. The Probabilistic Sharpe ratio PSR(0) was at least 0.99 in both universes, but the Sharpe advantage over classical baselines did not survive the Deflated Sharpe ratio.

Limitations and future work

Limitations. The evidence is limited to two equity-centric ETF universes at weekly frequency, under proportional transaction costs without market impact or borrowing costs. The filter relies on a Gaussian approximation to forward CVaR, which may understate losses when returns are fat-tailed, and it achieves compliance partly by holding portfolios close to minimum variance (mean L_1 distance 0.48 to 0.61). The constrained learner underperformed minimum variance in selloff weeks (Sharpe −0.809 against 0.035), its out-of-sample Sharpe advantage over classical baselines did not survive deflation, and the protocol reversal is documented on a single 31-asset window.

Future work. Replacing the Gaussian forecast with a semi-parametric or extreme-value estimate of tail risk would test whether breach depth can be reduced further. An adaptive dual update, such as the PID Lagrangian of [7], could remove the need to tune η_λ against the scale of the CVaR excess. Evaluation on daily data with market-impact costs, on non-equity universes and over multiple rolling start dates would establish how generally the breach reduction and the protocol reversal hold.

References. [1] R. T. Rockafellar, S. Uryasev. Optimization of conditional value-at-risk. *J. Risk* 2 (2000). [2] E. Altman. *Constrained Markov Decision Processes*. 1999. [3] G. Dalal et al. Safe exploration in continuous action spaces. arXiv:1801.08757, 2018. [4] D. H. Bailey, M. López de Prado. The deflated Sharpe ratio. *JPM* 40 (2014). [5] O. Ledoit, M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *JMVA* 88 (2004). [6] Y. Benjamini, Y. Hochberg. Controlling the false discovery rate. *JRSS B* 57 (1995). [7] A. Stooke, J. Achiam, P. Abbeel. Responsive safety in RL by PID Lagrangian methods. ICML 2020.

nl2992@columbia.edu yh3915@columbia.edu
github.com/nl2992/ICAIF_cvar-rl-portfolio-allocator